CHAPTER 2

Knots, Splices, Attachments, and Ladders

Section I. Knots, Hitches, and Lashings

A study of the terminology pictured in *Figure 2-1* and the definitions in *Table 2-1, page 2-2,* will aid in understanding the methods of knotting presented in this section.

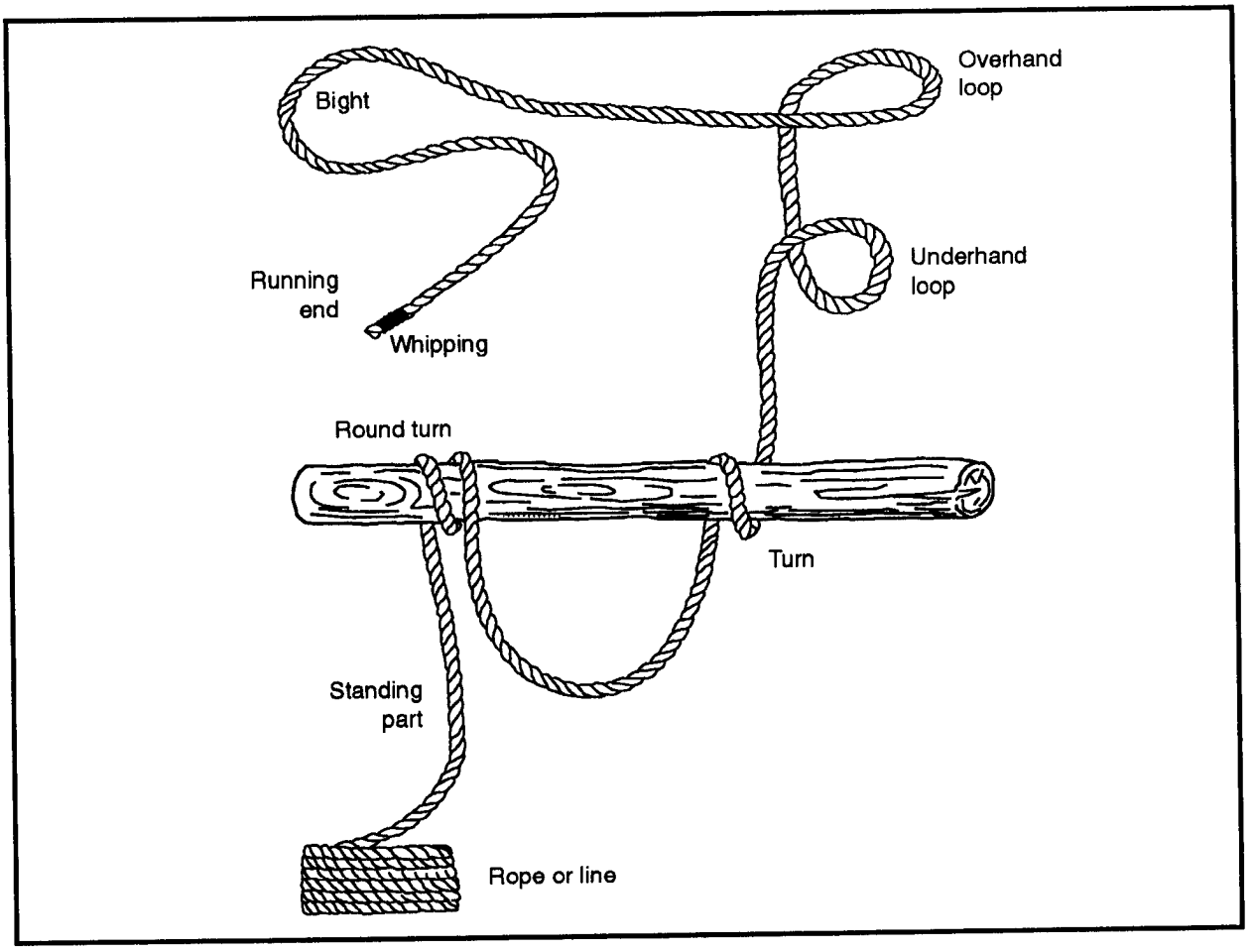


Figure 2-1. Elements of knots, bends, and hitches

Table 2-1. Knotting terminology and definitions

Term	Definition
Rope (often called a line)	A large, stout cord made of strands of fiber or wire that are twisted or braided together
Line (sometimes called a rope)	A thread, string, cord, or rope, especially a comparatively slender and strong cord (This manual will use the word rope rather than line in describing knots, hitches, rigging, and the like.)
Running end	The free or working end of a rope
Standing part	The rest of the rope, excluding the running end
Bight	A bend or U-shaped curve in a rope
Loop	Formed by crossing the running end over or under the standing part, forming a ring or circle in the rope
Turn	The placing of a loop around a specific object (such as a post, rail, or ring) with the running end continuing in a direction opposite to the standing part
Round turn	A modified turn, but with the running end leaving the circle in the same general direction as the standing part
Overhand turn or loop	Made when the running end passes over the standing part
Underhand turn or loop	Made when the running end passes under the standing part
Bend (in this manual, called a knot)	Used to fasten two ropes together or to fasten a rope to a ring or loop

The raw, cut end of a rope has a tendency to untwist and should always be knotted or fastened in some manner to prevent this untwisting. Whipping is one method of fastening the end of the rope to prevent untwisting (see *Figure 2-2*). A rope is whipped by wrapping the end tightly with a small cord. This method is particularly satisfactory because there is very little increase in the size of the rope. The whipped end of a rope will still thread through blocks or other openings. Before cutting a rope, place two whippings on the rope 1 or 2 inches apart and make the cut between the whippings (see *Figure 2-2*). This will prevent the cut ends from untwisting immediately after they are cut.

KNOTS

A knot is an interlacement of the parts of one or more flexible bodies, such as cordage rope, forming a lump. It is also any tie or fastening formed with a rope, including bends, hitches, and splices. A knot is often used as a stopper to prevent a rope from passing through an opening.

A good knot must be easy to tie, must hold without slipping, and must be easy to untie. The choice of the best knot, bend, or

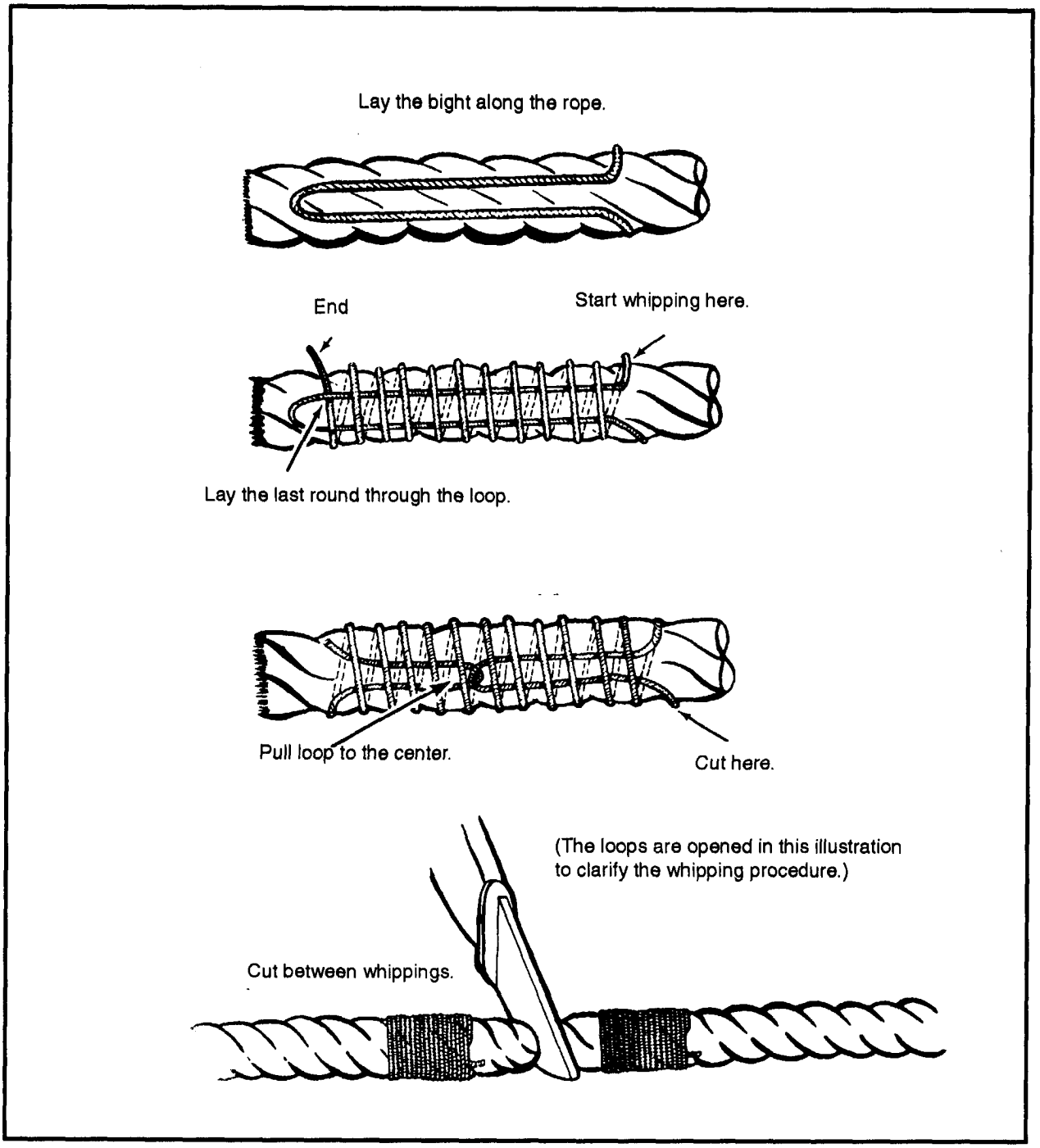


Figure 2-2. Whipping the end of a rope

hitch to use depends largely on the job it has to do. In general, knots can be classified into three groups. They are—

- Knots at the end of a rope.
- Knots for joining two ropes.
- Knots for making loops.

KNOTS AT THE END OF ROPE

Knots at the end of a rope fall into the following categories:

- Overhand knot.
- Figure-eight knot.
- Wall knot.

Overhand Knot

The overhand knot is the most commonly used and the simplest of all knots (see *Figure 2-3).* Use an overhand knot to prevent the end of a rope from untwisting, to form a knob at the end of a rope, or to serve as a part of another knot. When tied at the end or standing part of a rope, this knot prevents it from sliding through a block, hole, or another knot. Use it also to increase a person's grip on a rope. This knot reduces the strength of a straight rope by 55 percent.

Figure-Eight Knot

Use the figure-eight knot to form a larger knot at the end of a rope than would be formed by an overhand knot (see *Figure 2-4).* The knot prevents the end of the rope from slipping through a fastening or loop in another rope or from unreeving when reeved through blocks. It is easy to untie.

Wall Knot

Use the wall knot with crown to prevent the end of a rope from untwisting when an

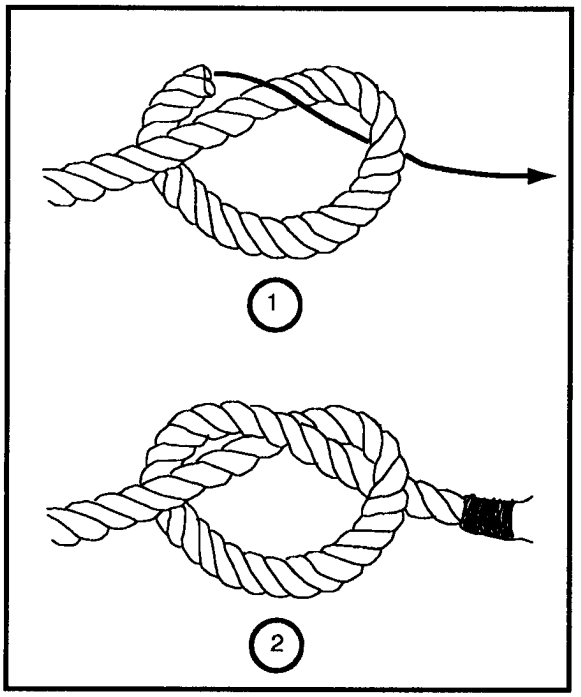


Figure 2-3. Overhand knot

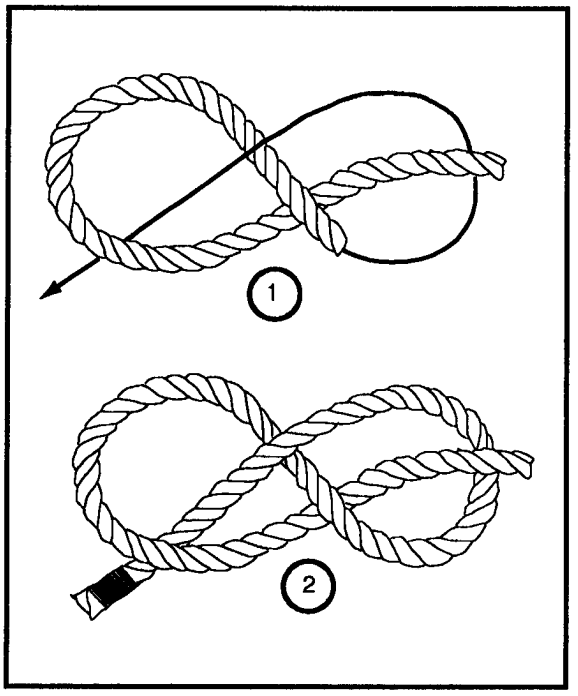


Figure 2-4. Figure-eight knot

enlarged end is not objectionable (see *Figure 2-5).* The wall knot also makes a desirable knot to prevent the end of the rope from slipping through small openings, as when using rope handles on boxes. Use either the crown or the wall knot separately to form semipermanent "stopper knots" tied with the end strands of a rope. The wall knot will prevent the rope from untwisting, but to make a neat round knob, crown it (see *Figure 2-6, page 2-6).* Notice that in the wall knot, the ends come up through the bights, causing the strands to lead forward. In a crown knot, the ends go down through the bights and point backward.

KNOTS FOR JOINING TWO ROPES

Knots for joining two ropes fall into the following categories:

- Square knot.
- Single sheet bend.
- Double sheet bend.
- Carrick bend.

Square Knot

Use the square knot to tie two ropes of equal size together so they will not slip (see

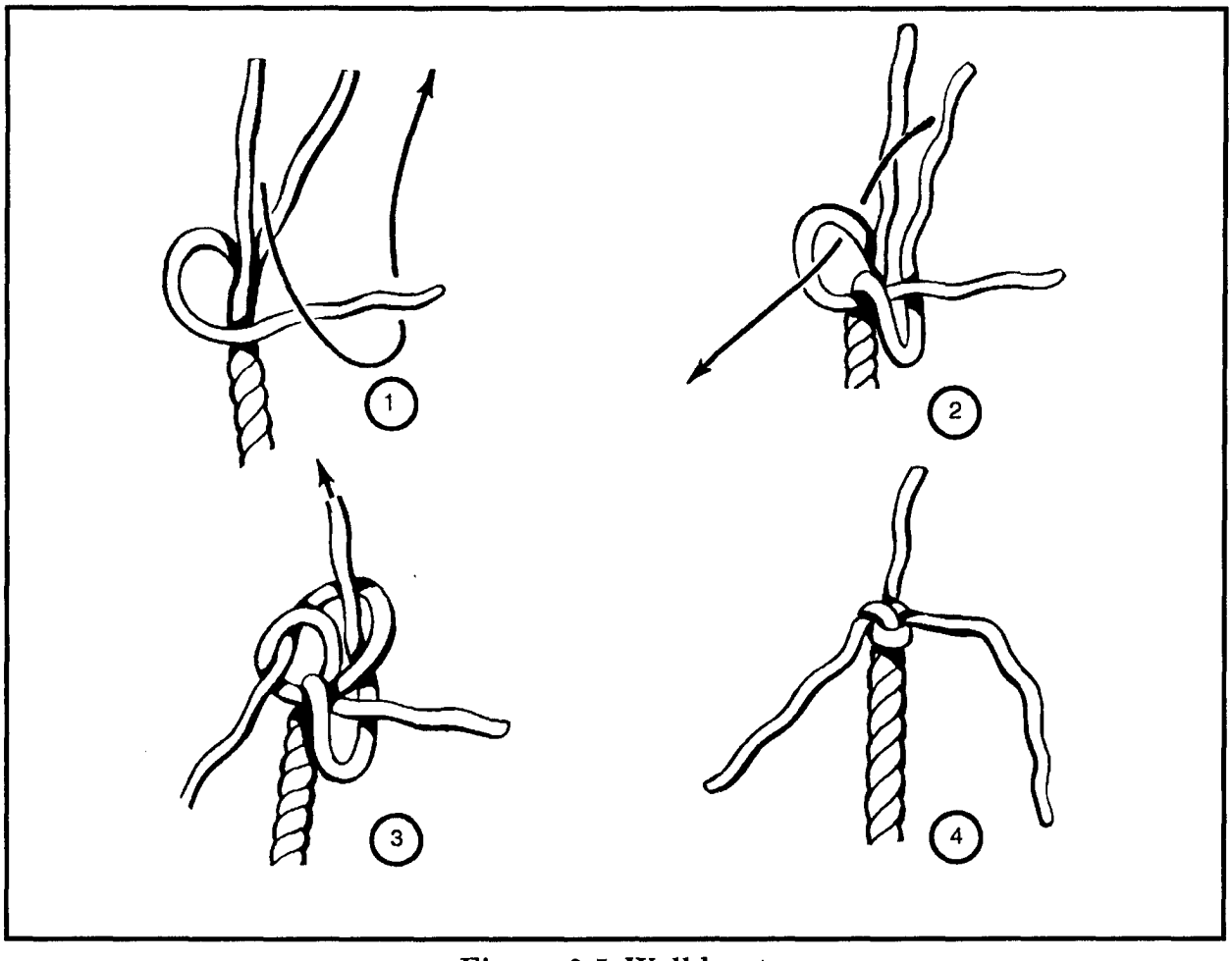


Figure 2-5. Wall knot

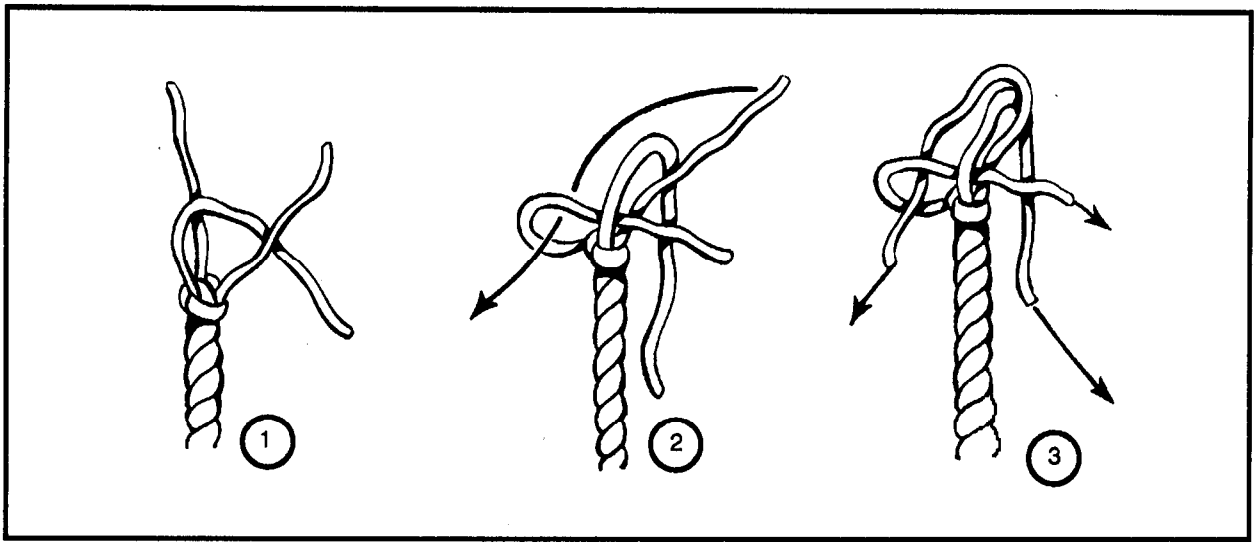


Figure 2-6. Crown on a wall knot

Figure 2-7). Note that in the square knot, the end and standing part of one rope come out on the same side of the bight formed by the other rope. The square knot will not hold if the ropes are wet or if they are of different sizes. It tightens under strain but can be untied by grasping the ends of the two bights and pulling the knot apart.

NOTE. **It makes no difference whether the first crossing is tied left-over-right or right-over-left as long as the second crossing is tied opposite to the first crossing.**

Single Sheet Bend

A single sheet bend, sometimes called a weaver's knot, has two major uses (see *Figure 2-8*). They are—

- Tying together two ropes of unequal size.
- Tying a rope to an eye.

This knot will draw tight but will loosen or slip when the lines are slackened. The single sheet bend is stronger and unties easier than the square knot.

Double Sheet Bend

The double sheet bend has greater holding power than the single sheet bend for joining ropes of equal or unequal diameter, joining wet ropes, or tying a rope to an eye (see *Figure 2-9, page 2-8,).* It will not slip or draw tight under heavy loads. This knot is more secure than the single sheet bend when used in a spliced eye.

Carrick Bend

Use the carrick bend for heavy loads and for joining large hawsers or heavy rope (see *Figure 2-10, page 2-8).* It will not draw tight under a heavy load and can be untied easily if the ends are seized to their own standing part.

KNOTS FOR MAKING LOOPS

Knots for making loops fall into the following categories:

- Bowline.
- Double bowline.
- Running bowline.

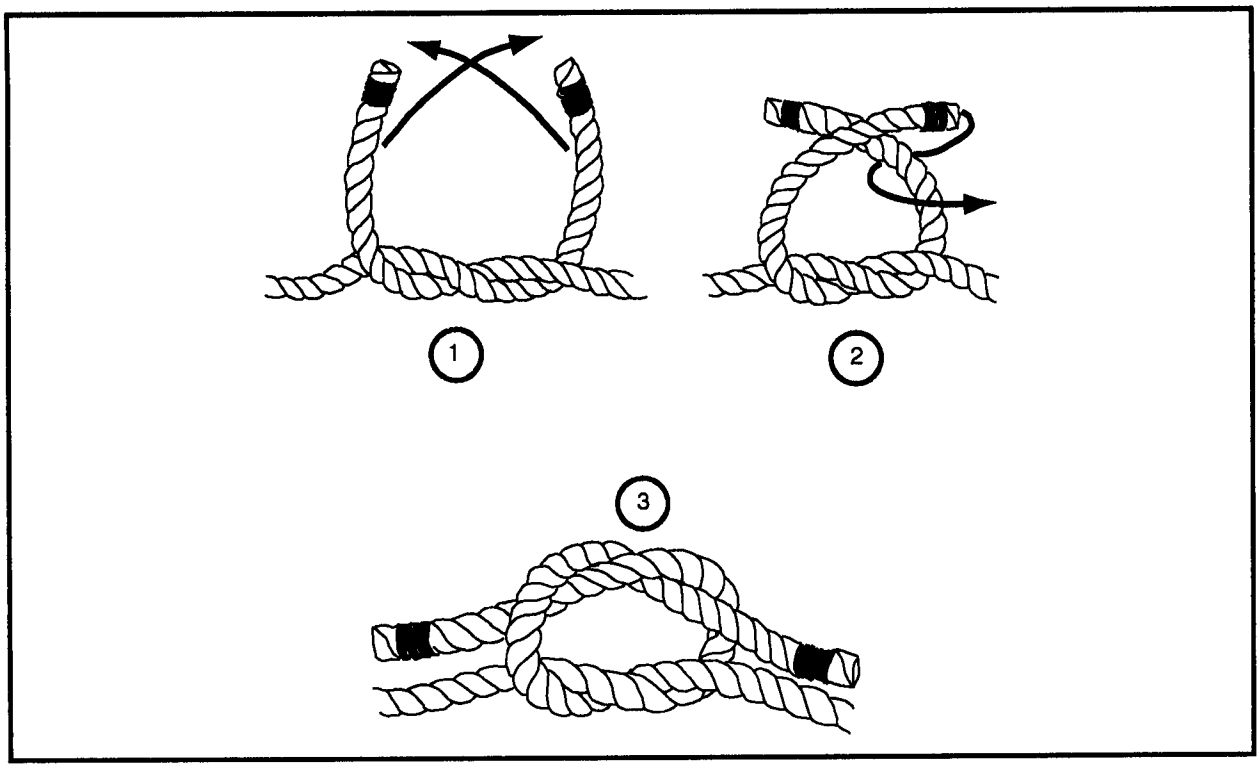


Figure 2-7. Square knot

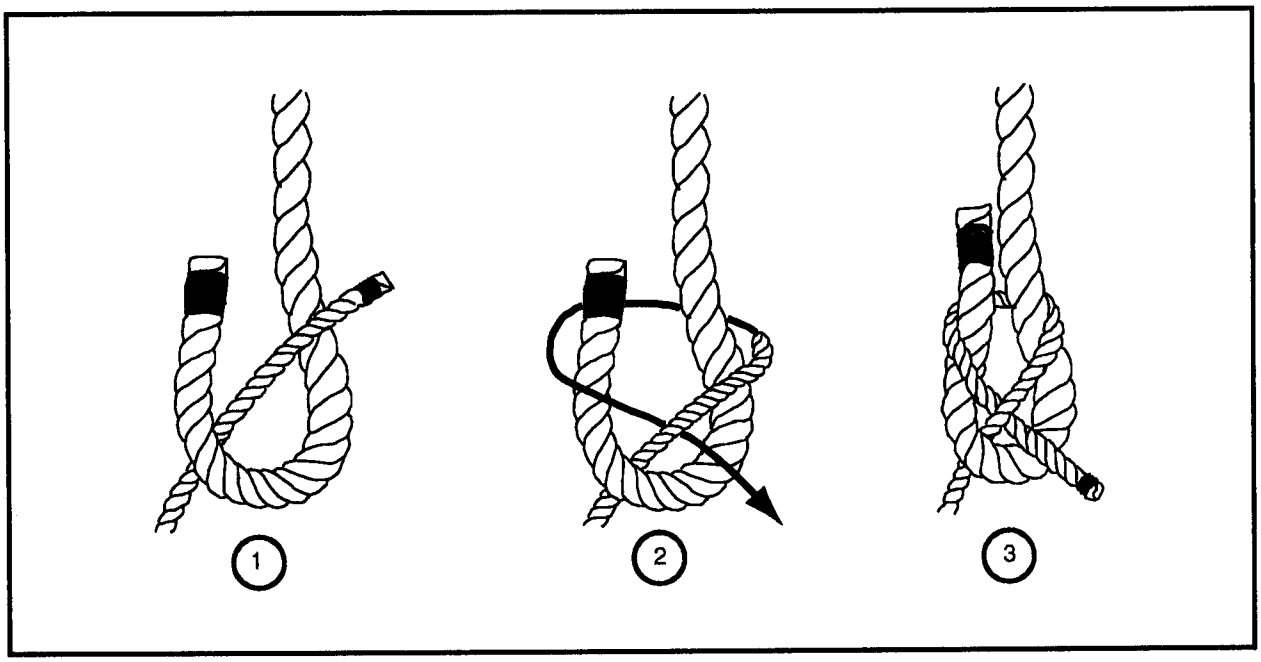


Figure 2-8. Single sheet bend

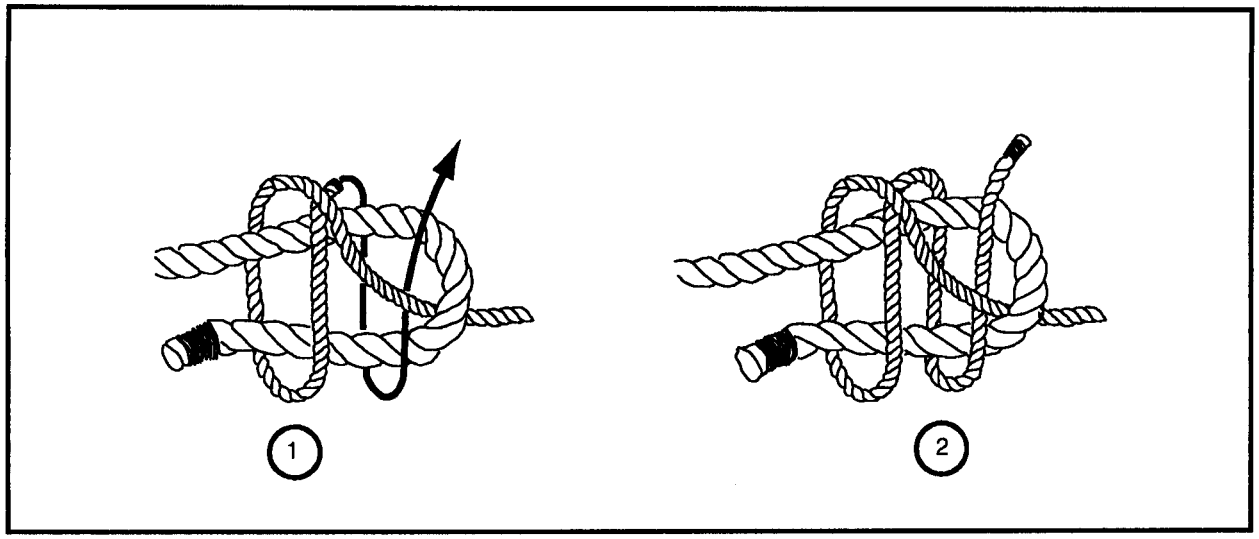


Figure 2-9. Double sheet bend

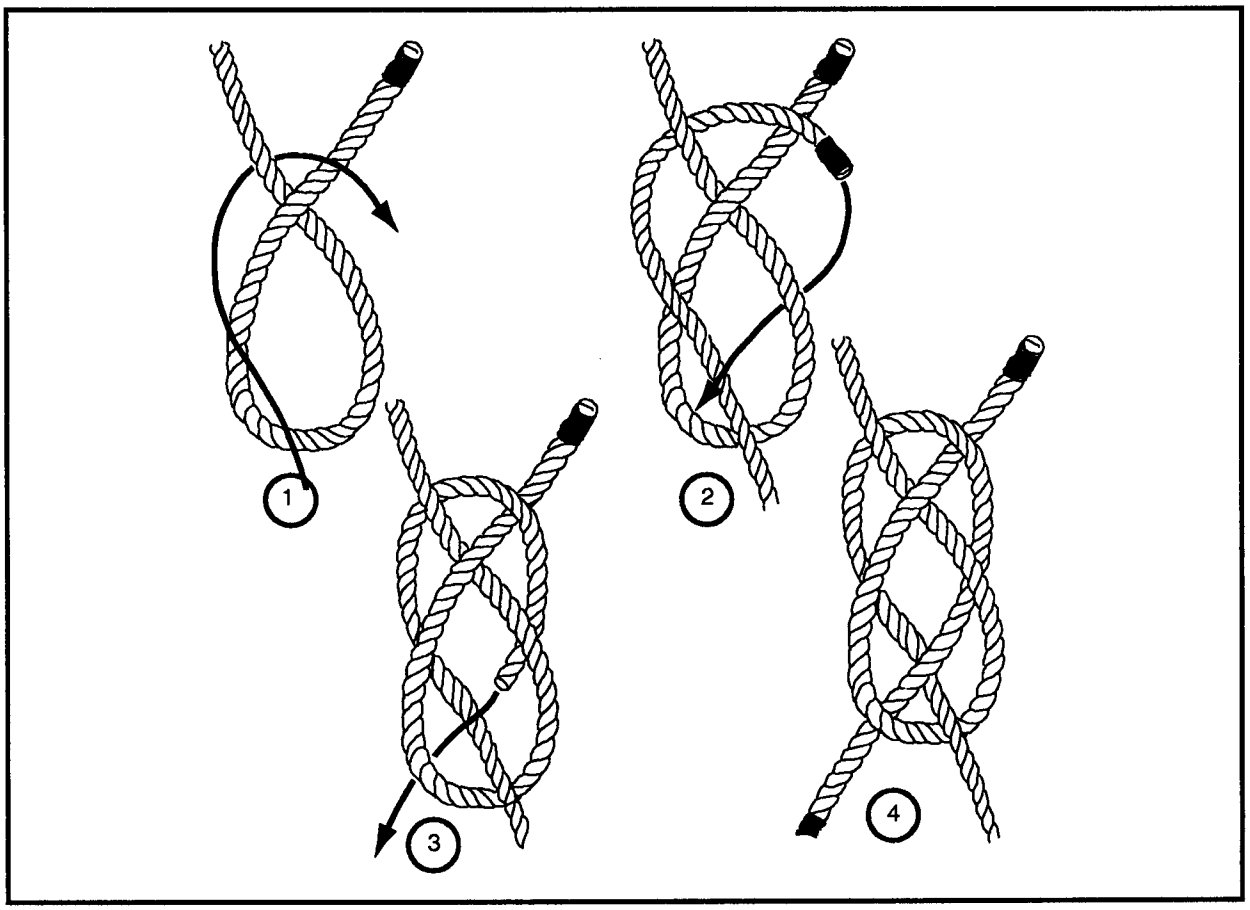


Figure 2-10. Carrick bend

- Bowline on a bight.
- Spanish bowline.
- French bowline.
- Speir knot.
- Cat's-paw.
- Figure eight with an extra turn.

Bowline

The bowline is one of the most common knots and has a variety of uses, one of which is the lowering of men and material (see *Figure 2-11).* It is the best knot for forming a single loop that will not tighten or slip under strain and can be untied easily if each running end is seized to its own standing part. The bowline forms a loop that may be of any length.

Double Bowline

The double bowline forms three nonslipping loops (see *Figure 2-12, page 2-10).* Use this knot to sling a man. As he sits in the slings, one loop supports his back and the remaining two loops support his legs. A notched board that passes through the two loops makes a comfortable seat known as a boatswain's chair. This chair is discussed in the scaffolding section of this manual (see *Chapter 6).*

Running Bowline

The running bowline forms a strong running loop (see *Figure 2-13, page 2-10).* It is a convenient form of running an eye. The running bowline provides a sling of the choker type at the end of a single line. Use it when tying a handline around an object at a point that you cannot safely reach, such as the end of a limb.

Bowline on a Bight

This knot forms two nonslipping loops (see *Figure 2-14, page 2-11).* You can use the bowline on a bight for the same purpose as a boatswain's chair. It does not leave both hands free, but its twin nonslipping loops form a comfortable seat. Use it when—

- You need more strength than a single bowline will give.

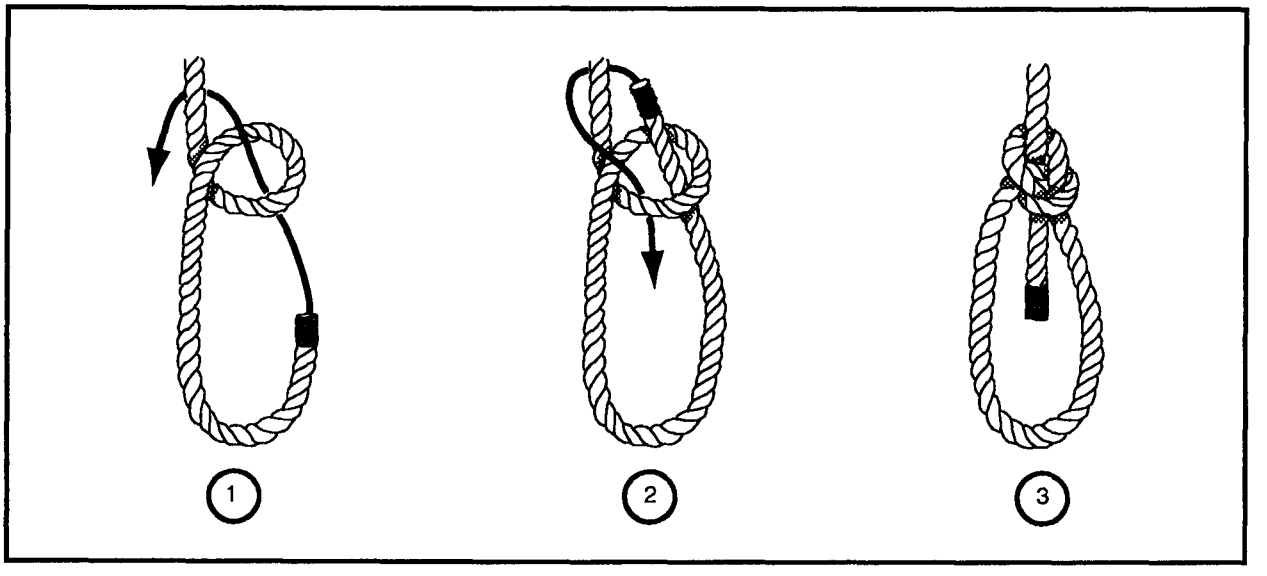


Figure 2-11. Bowline

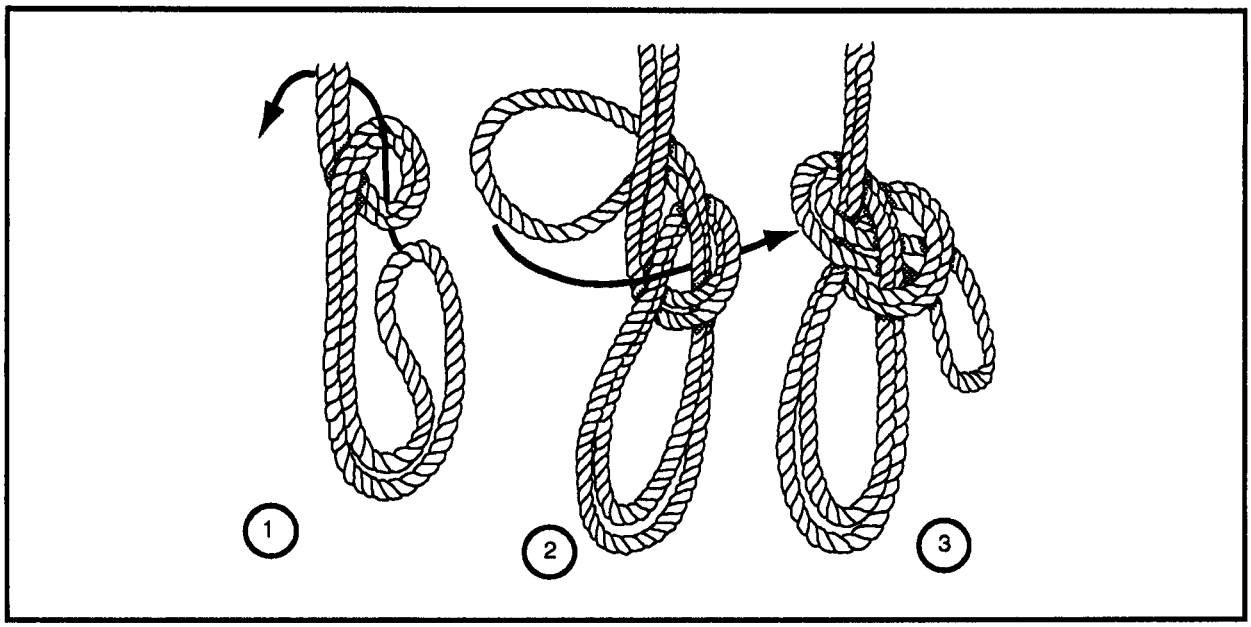


Figure 2-12. Double bowline

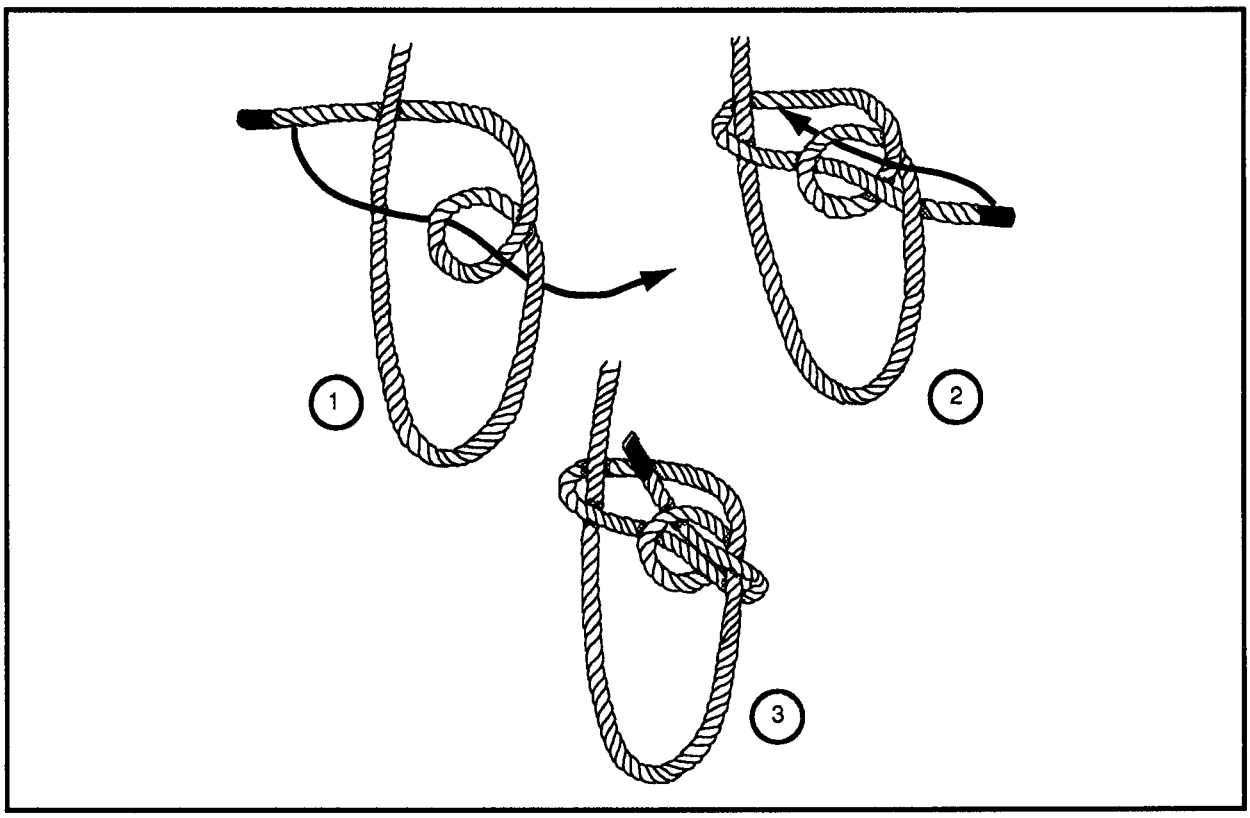


Figure 2-13. Running bowline

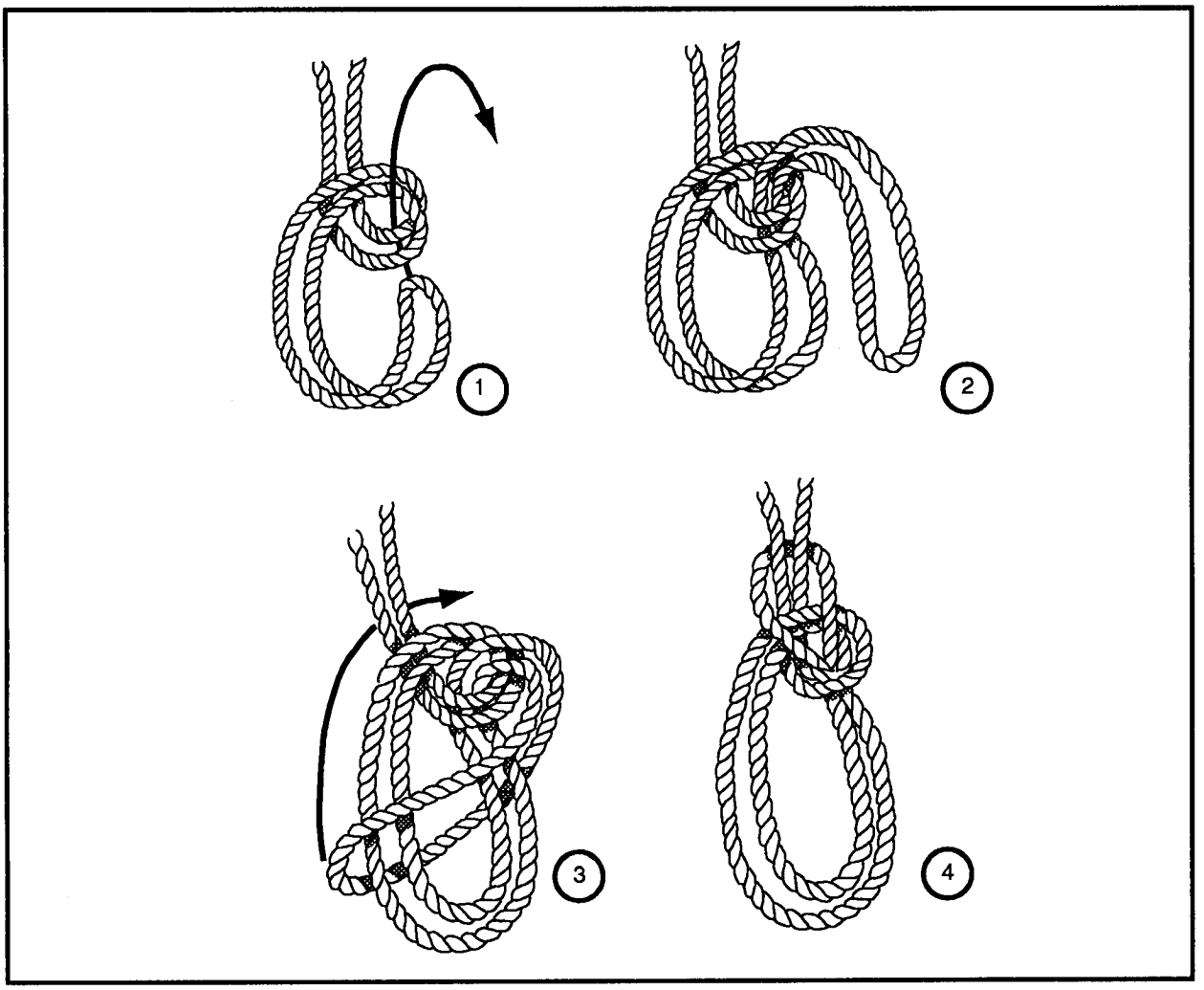


Figure 2-14. Bowline on a bight

- You need to form a loop at some point in a rope other than at the end.
- You do not have access to the end of a rope.

You can easily untie the bowline on a bight and tie it at the end of a rope by doubling the rope for a short section.

Spanish Bowline

You can tie a Spanish bowline at any point in a rope, either at a place where the line is double or at an end that has been doubled back (see *Figure 2-15, page 2-12).* Use the Spanish bowline in rescue work or to give a twofold grip for lifting a pipe or other round objects in a sling.

French Bowline

You can use the French bowline as a sling to lift injured men (see *Figure 2-16, page 2-12).* When used for this purpose, one loop is a seat and the other loop is put around the body under the arms. The injured man's weight keeps the two loops tight so that he cannot fall out. It is particularly useful as a sling for an unconscious man. Also, use the French bowline when working alone and you

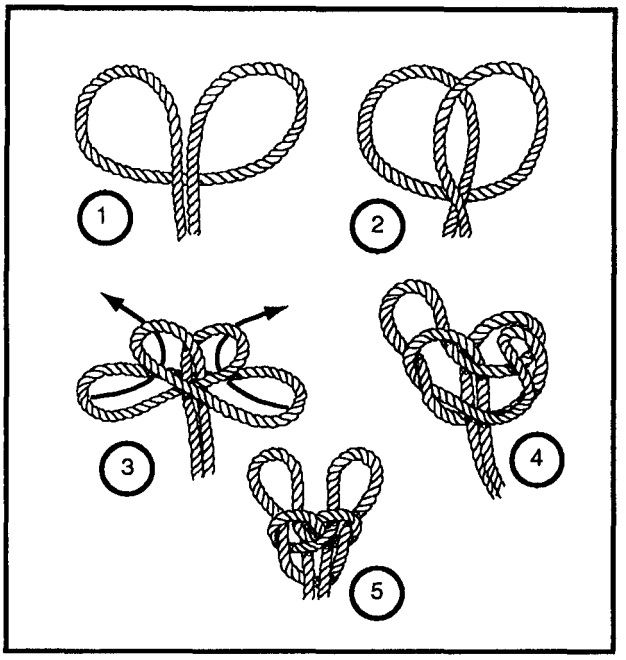


Figure 2-15. Spanish bowline

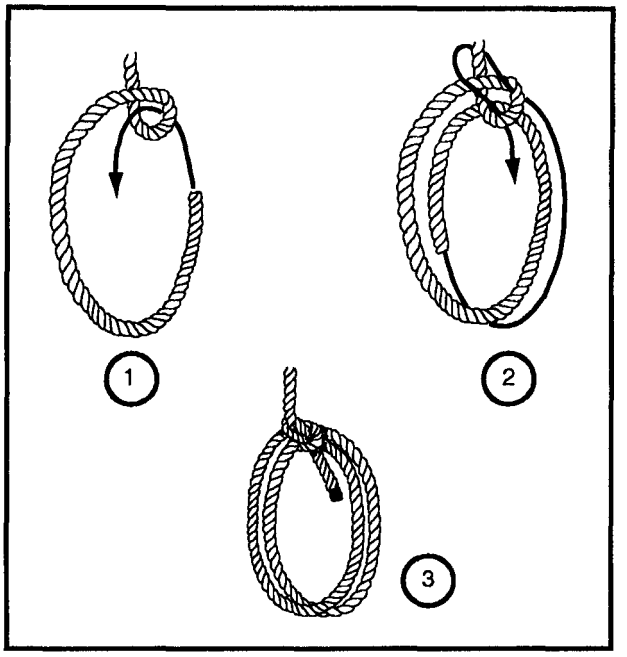


Figure 2-16. French bowline

need your hands free. The two loops of this knot can be adjusted to the size required.

Speir Knot

Use a speir knot when you need a fixed loop, a nonslip knot, and a quick release (see *Figure 2-17*). You can tie this knot quickly and release it by pulling on the running end.

Cat's-paw

Use a cat's-paw to fasten an endless sling to a hook, or make it at the end of a rope to fasten the rope to a hook (see *Figure 2-18*). You can tie or untie it easily. This knot, which is really a form of a hitch, is a more satisfactory way of attaching a rope to a hook than the blackwall hitch. It will not slip off and need not be kept taut to make it hold.

Figure Eight With an Extra Turn

Use a figure eight with an extra turn to tighten a rope (see *Figure 2-19, page 2-14).* This knot is especially suitable for tightening a one-rope bridge across a small stream. You can tie and untie it easily.

KNOTS FOR TIGHTENING A ROPE

The types of knots used for tightening a rope are the butterfly knot and the baker bowline.

Butterfly Knot

Use the butterfly knot is to pull taut a high line, handline, tread rope for foot bridges, or similar installations (see *Figure 2-20, page 2-14).* Using this knot provides the capability to tighten a fixed rope when mechanical means are not available. (You can also use the harness hitch for this purpose [see *Figure 2-32, page 2-22]).* The butterfly knot will not jam if a stick is placed between the two upper loops.

Baker Bowline

You can use the baker bowline for the same purpose as the butterfly knot and for lashing cargo (see *Figure 2-21, pages 2-15 and 2-16).*

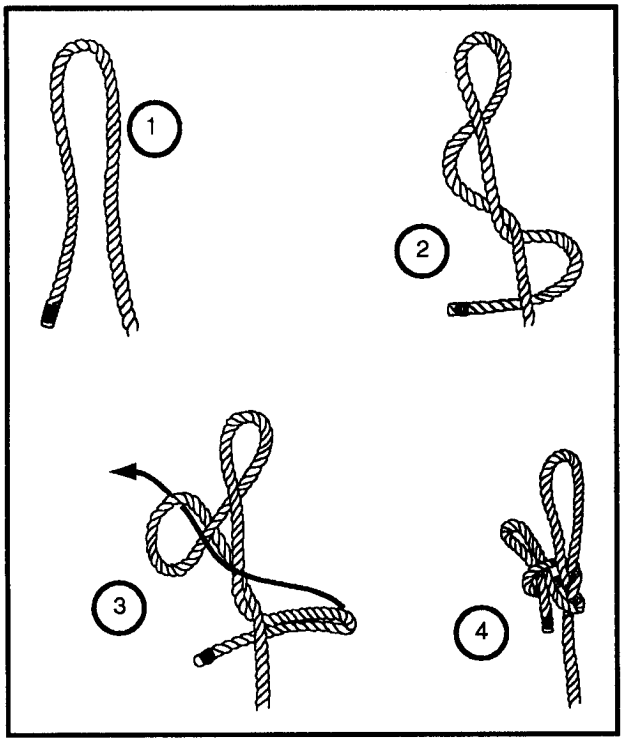


Figure 2-17. Speir knot

When used to lash cargo, secure one end with two half hitches, pass the rope over the cargo and tie a baker bowline, then secure the lashing with a slippery half hitch. To release the rope, simply pull on the running end. Advantages of the baker bowline are that it can be—

- Tied easily.
- Adjusted without losing control.
- Released quickly.

KNOTS FOR WIRE ROPE

Under special circumstances, when wire-rope fittings are not available and it is necessary to fasten wire rope by some other manner, you can use certain knots. In all knots made with wire rope, fasten the running end of the rope to the standing part after tying the knot. When wire-rope clips are available, use them to fasten the running end. If clips are not available, use

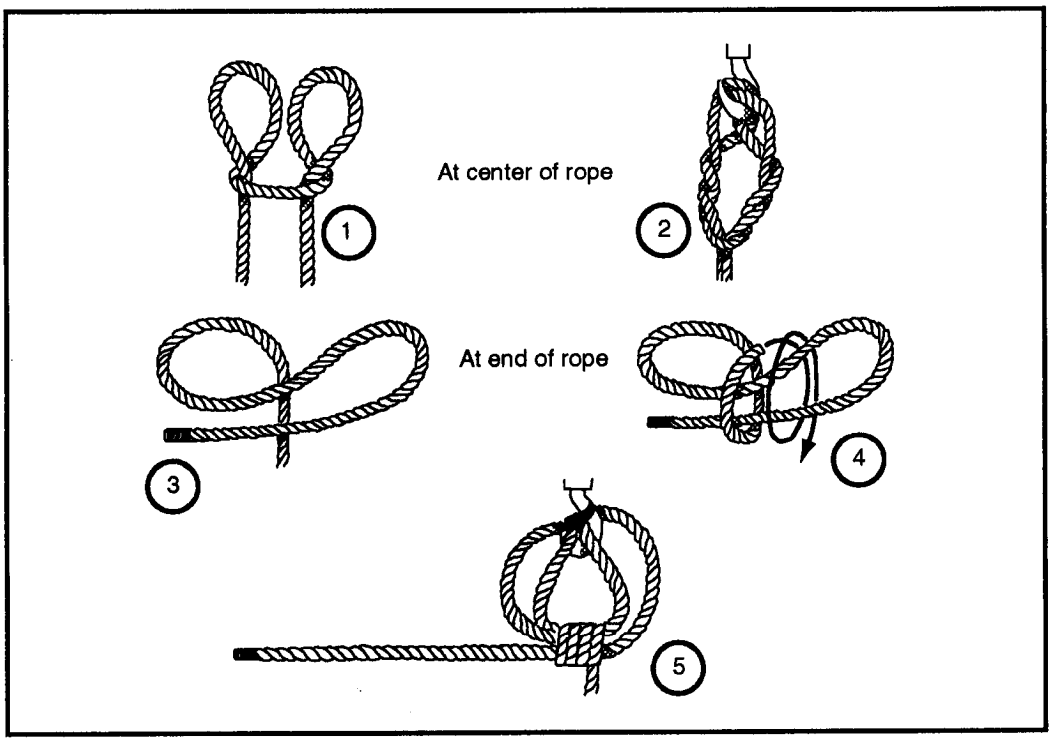


Figure 2-18. Cats-paw

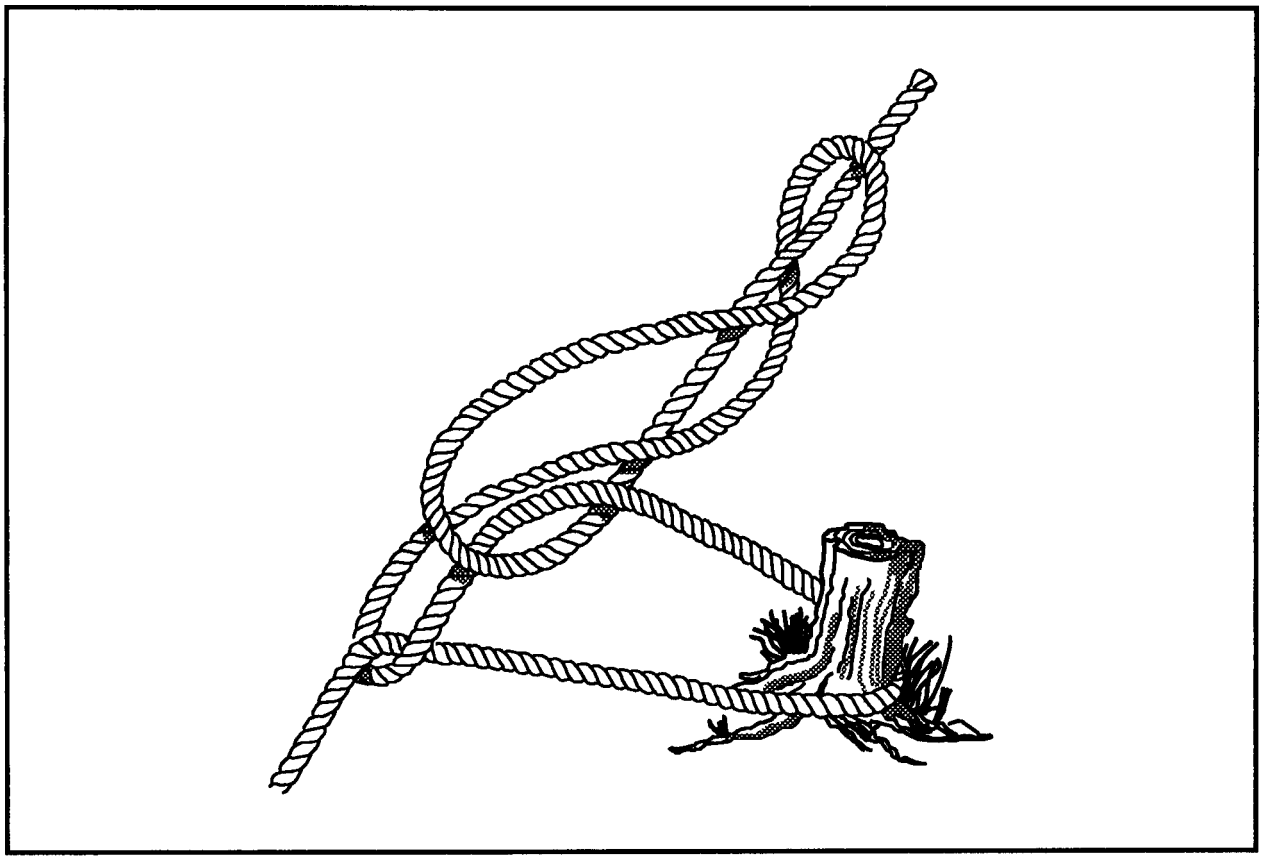


Figure 2-19. Figure eight with an extra turn

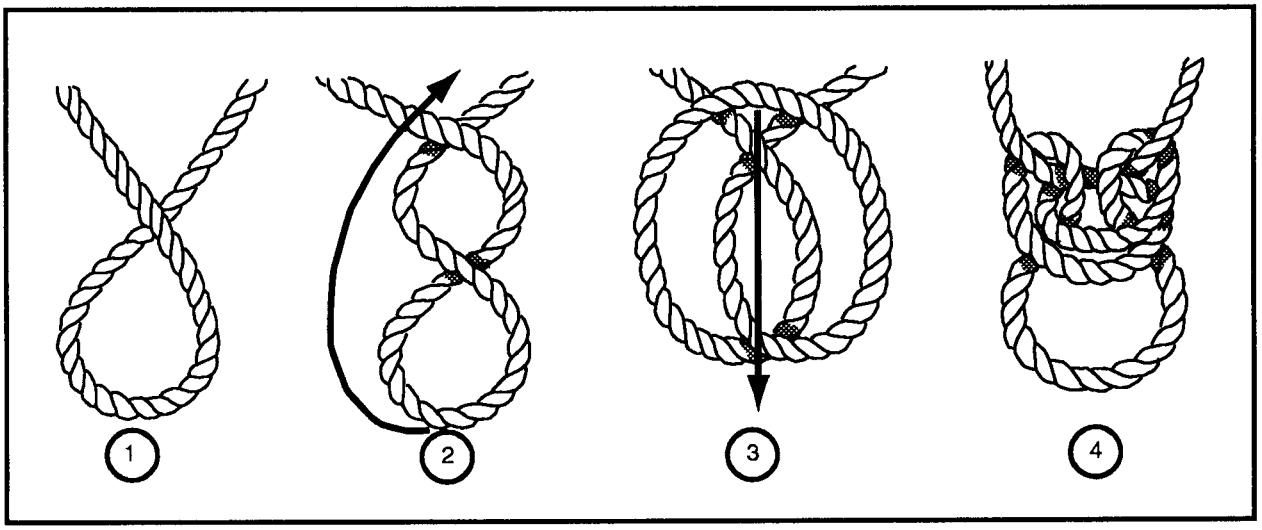


Figure 2-20. Butterfly knot

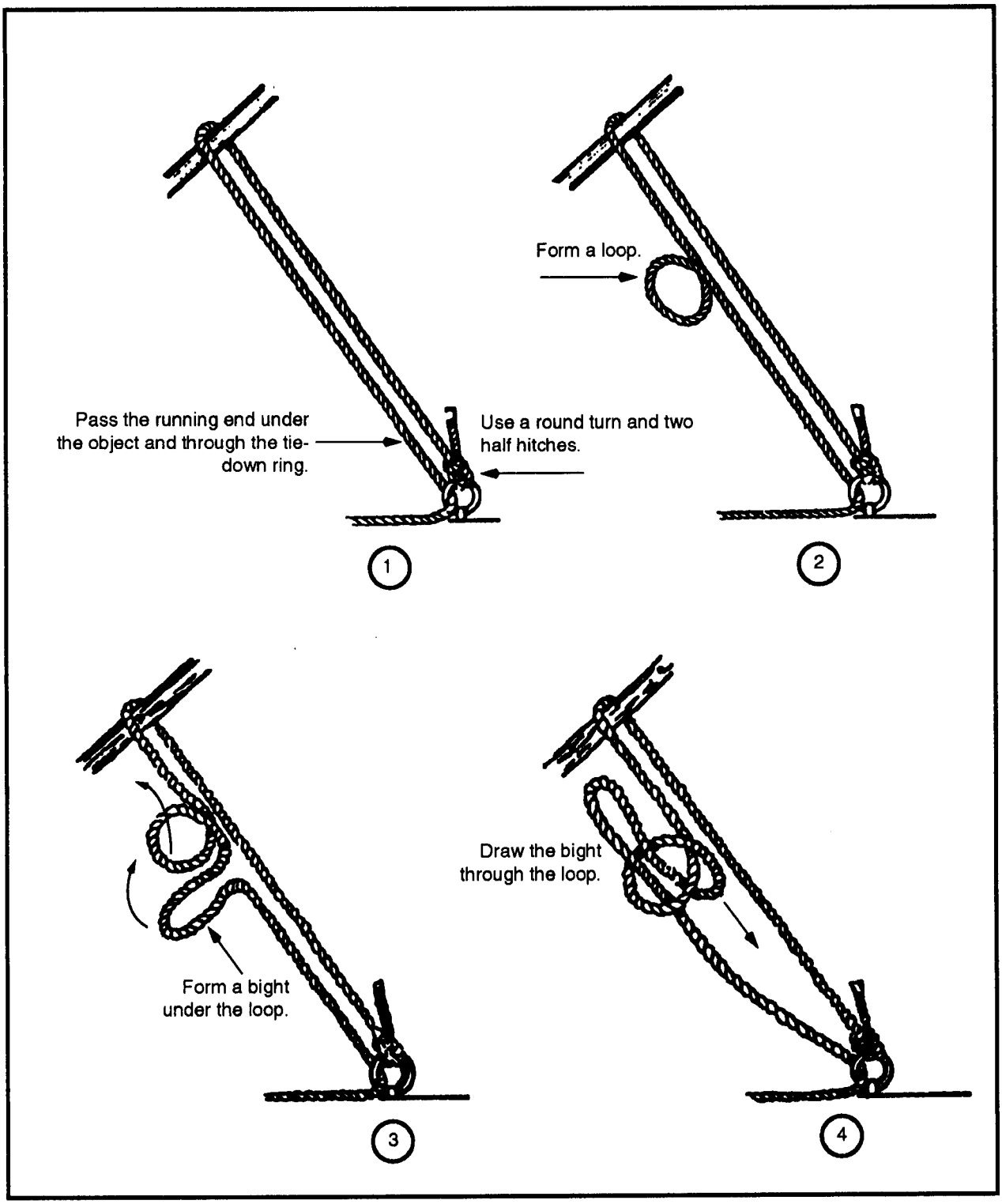


Figure 2-21. Baker bowline

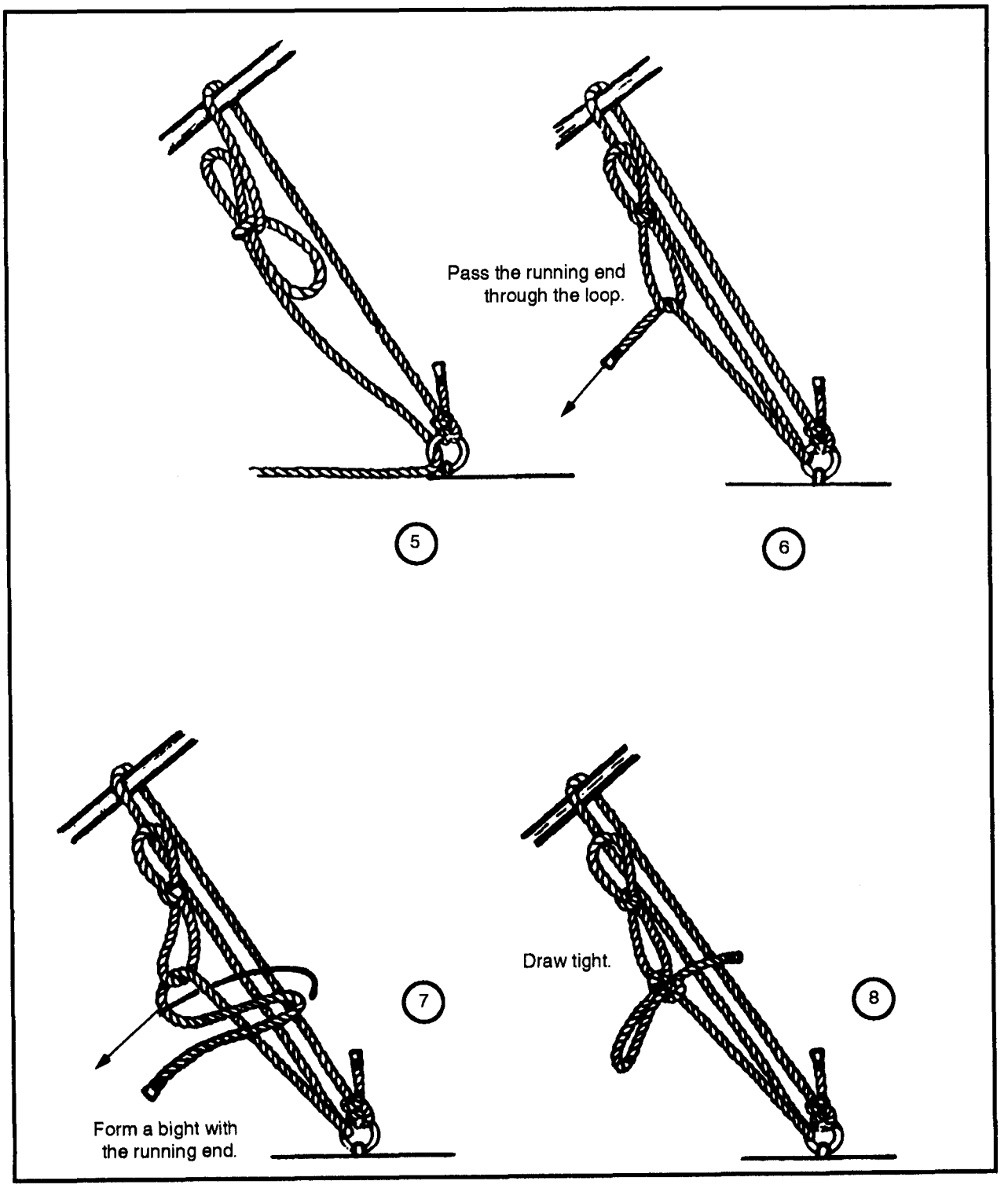


Figure 2-21. Baker bowline (continued)

wire or strands of cordage. Check all knots in wire rope periodically for wear or signs of breakage. If there is any reason to believe that the knot has been subjected to excessive wear, cut off a short length of the end of the rope, including the knot, and tie a new knot. Use the fisherman's bend, clove hitch, and carrick bend to fasten wire rope.

HITCHES

A hitch is any of various knots used to form a temporary noose in a rope or to secure a rope around a timber, pipe, or post so that it will hold temporarily but can be readily undone. The types of hitches are as follows:

- Half hitch.
- Two half hitches.
- Round turn and two half hitches.
- Timber hitch.
- Timber hitch and half hitch.
- Clove hitch.
- Rolling hitch.
- Telegraph hitch.
- Mooring hitch.
- Scaffold hitch.
- Blackwall hitch.
- Harness hitch.
- Girth hitch.
- Sheepshank.
- Fisherman's bend.

HALF HITCH

Use the half hitch to tie a rope to a timber or to a larger rope (see *Figure 2-22, A).* It will hold against a steady pull on the standing part of the rope; however, it is not a secure hitch. You can use the half hitch to secure the free end of a rope and as an aid to and the foundation of many knots. For example, it is the start of a timber hitch and a part of the fisherman's knot. It also makes the rolling hitch more secure.

TWO HALF HITCHES

Two half hitches are especially useful for securing the running end of a rope to the standing part (see *Figure 2-22, B).* If the two hitches are slid together along the standing part to form a single knot, the knot becomes a clove hitch.

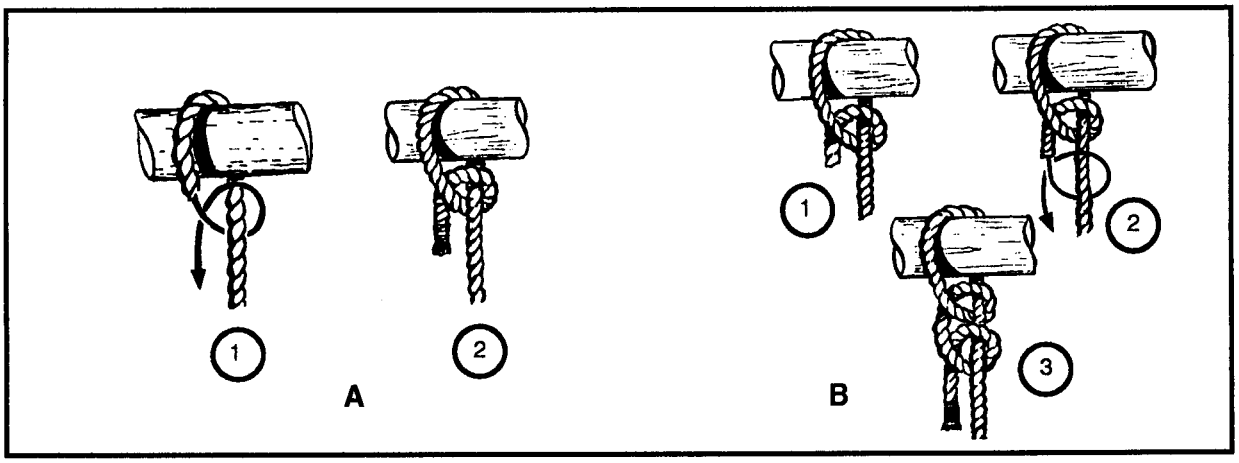


Figure 2-22. Half hitches

ROUND TURN AND TWO HALF HITCHES

Another hitch used to fasten a rope to a pole, timber, or spar is the round turn and two half hitches (see *Figure 2-23).* For greater security, seize the running end of the rope to the standing part. This hitch does not jam.

TIMBER HITCH

Use the timber hitch to move heavy timber or poles (see *Figure 2-24).* It is excellent for securing a piece of lumber or similar objects. The pressure of the coils, one over the other, holds the timber securely; the more tension applied, the tighter the hitch becomes about the timber. It will not slip but will readily loosen when the strain is relieved.

TIMBER HITCH AND HALF HITCH

A timber hitch and half hitch are combined to hold heavy timber or poles when they are being lifted or dragged (see *Figure 2-25).* A timber hitch used alone may become untied when the rope is slack or when a sudden strain is put on it.

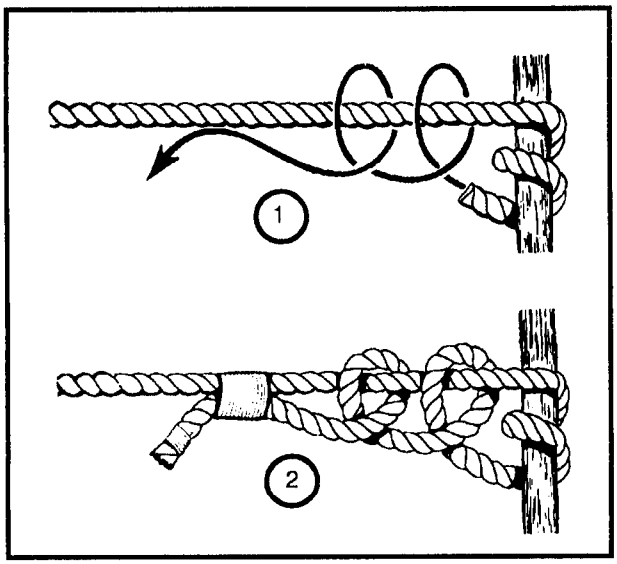


Figure 2-23. Round turn and two half hitches

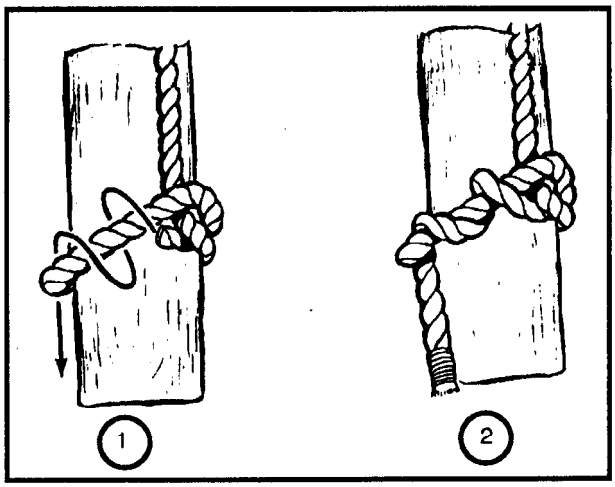


Figure 2-24. Timber hitch

CLOVE HITCH

The clove hitch is one of the most widely used knots (see *Figure 2-26, page 2-19).* You can use it to fasten a rope to a timber, pipe, or post. You can also use it to make other knots. This knot puts very little strain on the fibers when the rope is put around an object in one continuous direction. You can tie a clove hitch at any point in a rope. If there is not constant tension on the rope, another loop (round of the rope around the object and under the center of the clove hitch) will permit a tightening and slackening motion of the rope.

ROLLING HITCH

Use the rolling hitch to secure a rope to another rope or to fasten it to a pole or pipe

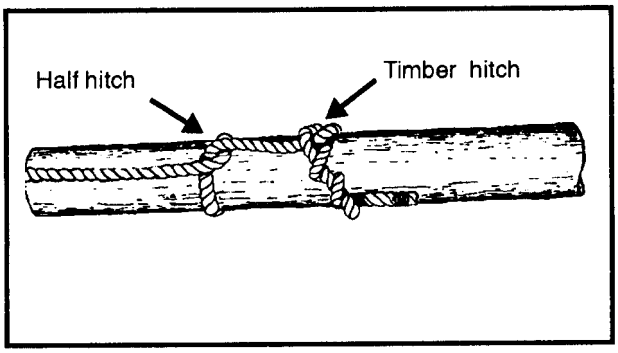


Figure 2-25. Timber hitch and half hitch

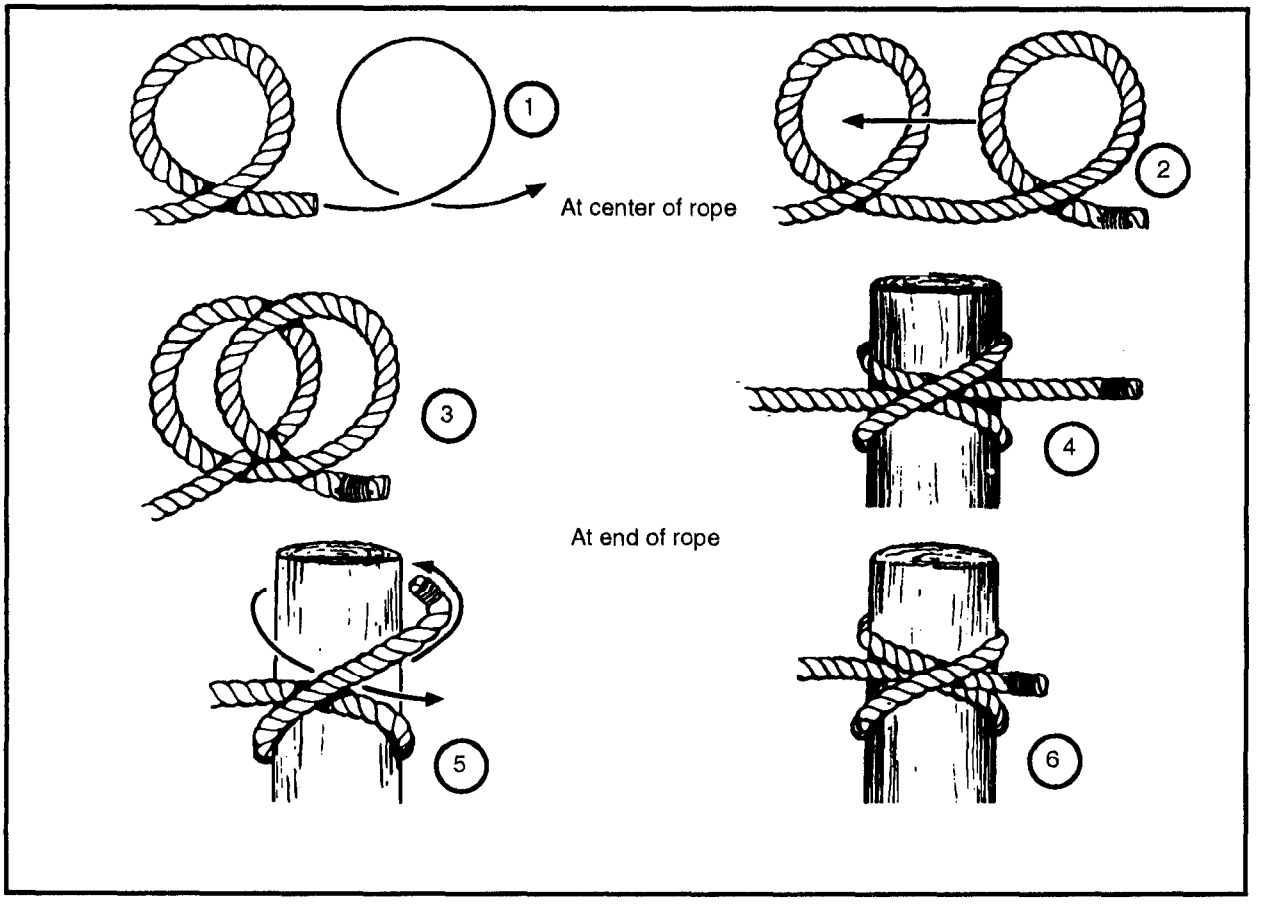


Figure 2-26. Clove hitch

so that the rope will not slip (see *Figure 2-27, page 2-20).* This knot grips tightly but is easily moved along a rope or pole when the strain is relieved.

TELEGRAPH HITCH

The telegraph hitch is a very useful and secure hitch that you can use to hoist or haul posts and poles (see *Figure 2-28, page 2-20).* It is easy to tie and untie and will not slip.

MOORING HITCH

Use the mooring hitch, also called rolling or magnus hitch, to fasten a rope around a mooring post or to attach a rope at a right angle to a post (see *Figure 2-29, page 2-21).* This hitch grips tightly and is easily removed.

SCAFFOLD HITCH

Use the scaffold hitch to support the end of a scaffold plank with a single rope (see *Figure 2-30, page 2-21).* It prevents the plank from tilting.

BLACKWALL HITCH

Use the blackwall hitch to fasten a rope to a hook (see *Figure 2-31, page 2-22).* Generally, use it to attach a rope, temporarily, to a hook or similar object in derrick work. The hitch holds only when subjected to a constant strain or when used in the middle of a

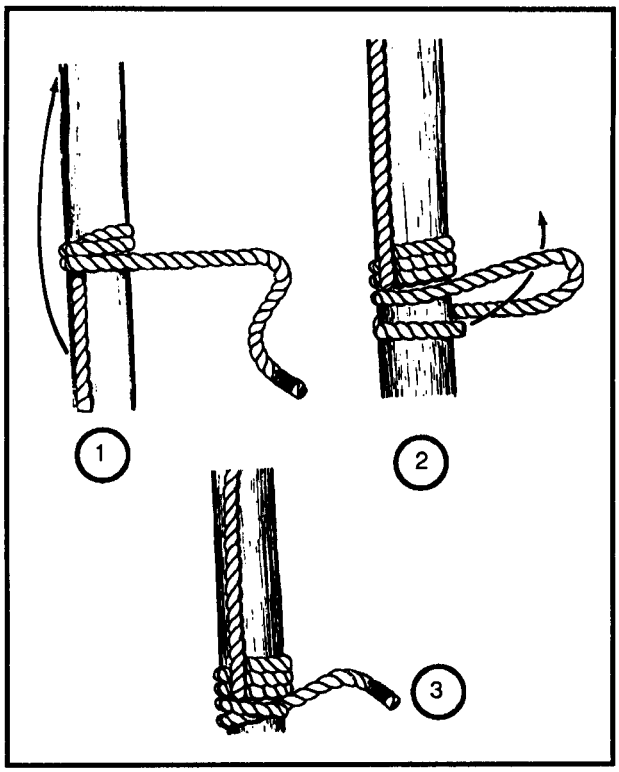


Figure 2-27. Rolling hitch

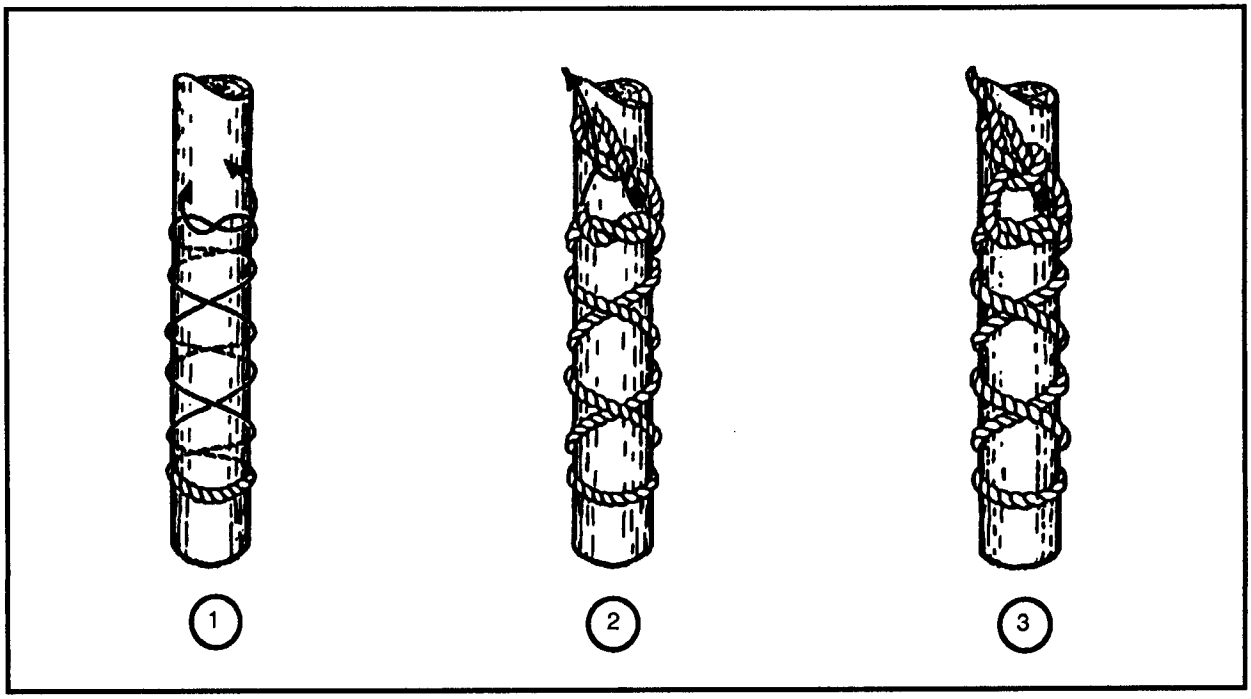


Figure 2-28. Telegraph hitch

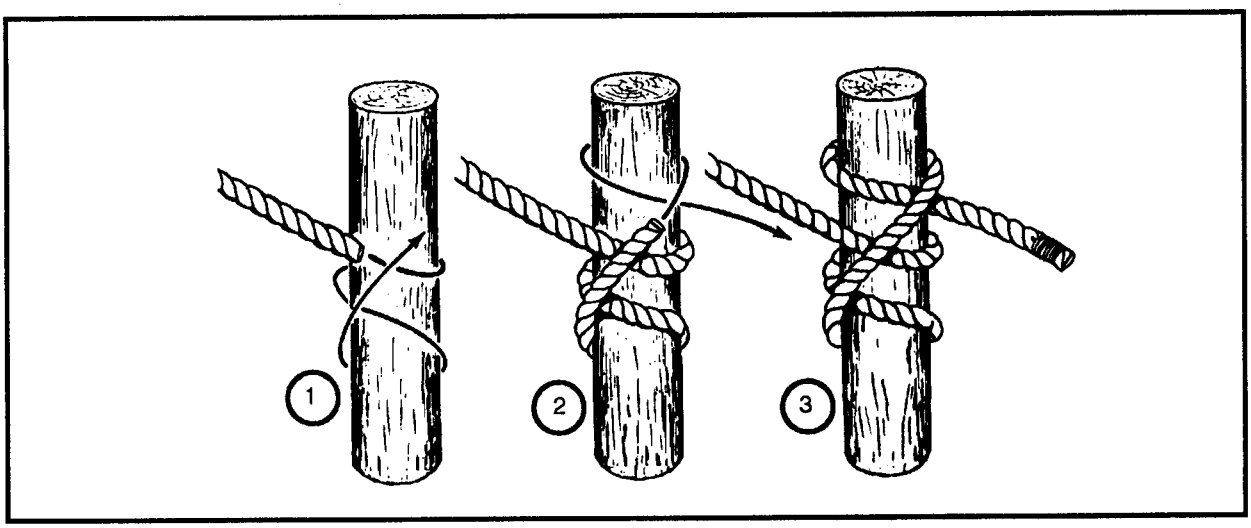


Figure 2-29. Mooring hitch

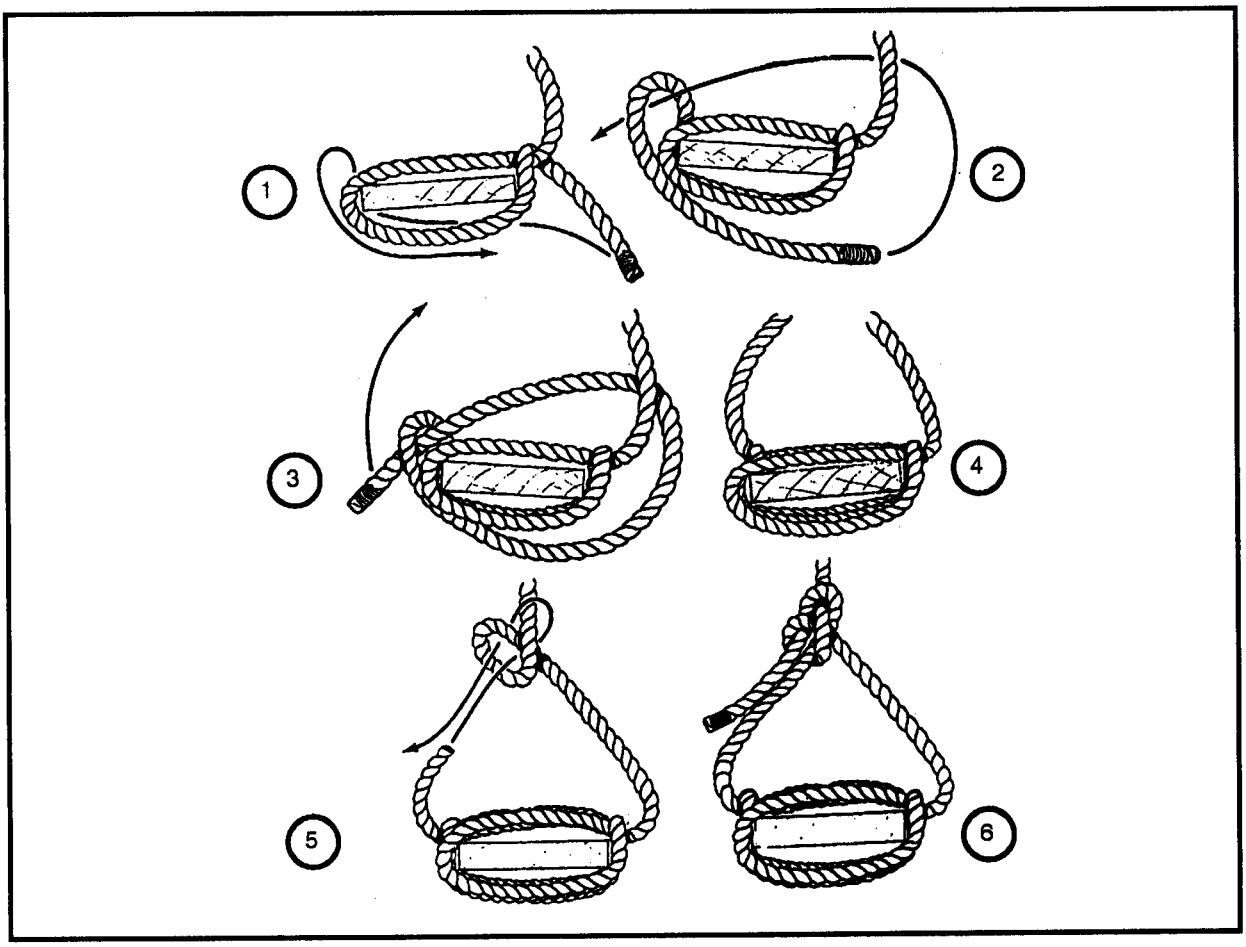


Figure 2-30. Scaffold hitch

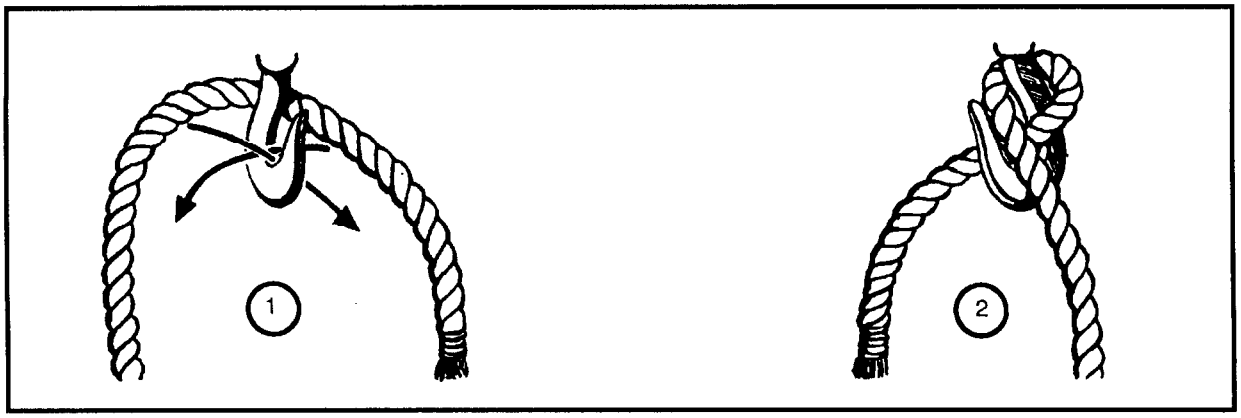


Figure 2-31. Blackwall hitch

rope with both ends secured. Human life and breakable equipment should never be entrusted to the blackwall hitch.

HARNESS HITCH

The harness hitch forms a nonslipping loop in a rope (see *Figure 2-32).* It is often employed by putting an arm through the loop, then placing the loop on the shoulder and pulling the object attached to the rope.

The hitch is tied only in the middle of a rope. It will slip if only one end of the rope is pulled.

GIRTH HITCH

Use the girth hitch to tie suspender ropes to hand ropes when constructing expedient foot bridges (see *Figure 2-33).* It is a simple and convenient hitch for many other uses of ropes and cords.

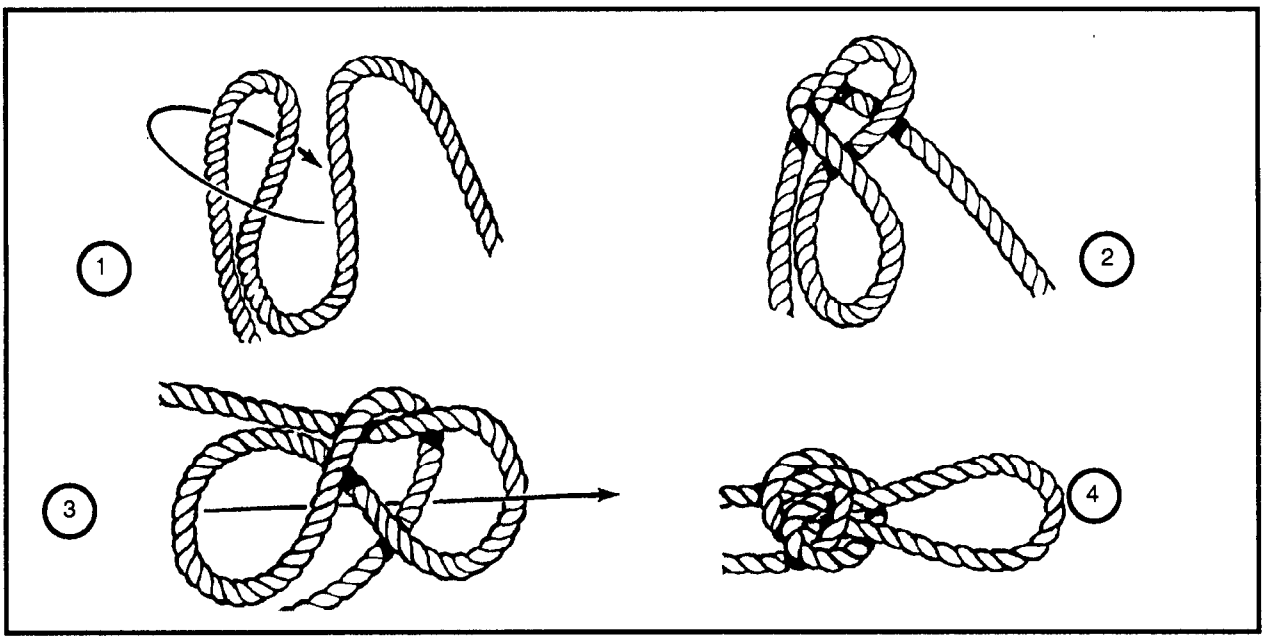


Figure 2-32. Harness hitch

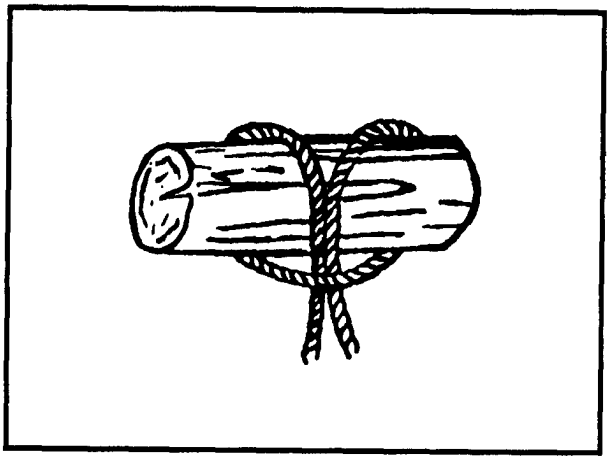


Figure 2-33. Girth hitch

SHEEPSHANK

A sheepshank is a method of shortening a rope, but you can use it to take the load off a weak spot in the rope (see *Figure 2-34).* It is only a temporary knot unless the eyes are fastened to the standing part on each end.

FISHERMAN'S BEND

The fisherman's bend is an excellent knot for attaching a rope to a light anchor, a ring, or a rectangular piece of stone (see *Figure 2-35, page 2-24).* You can use it to fasten a rope or cable to a ring or post. Also use it where there will be a slackening and tightening motion in the rope.

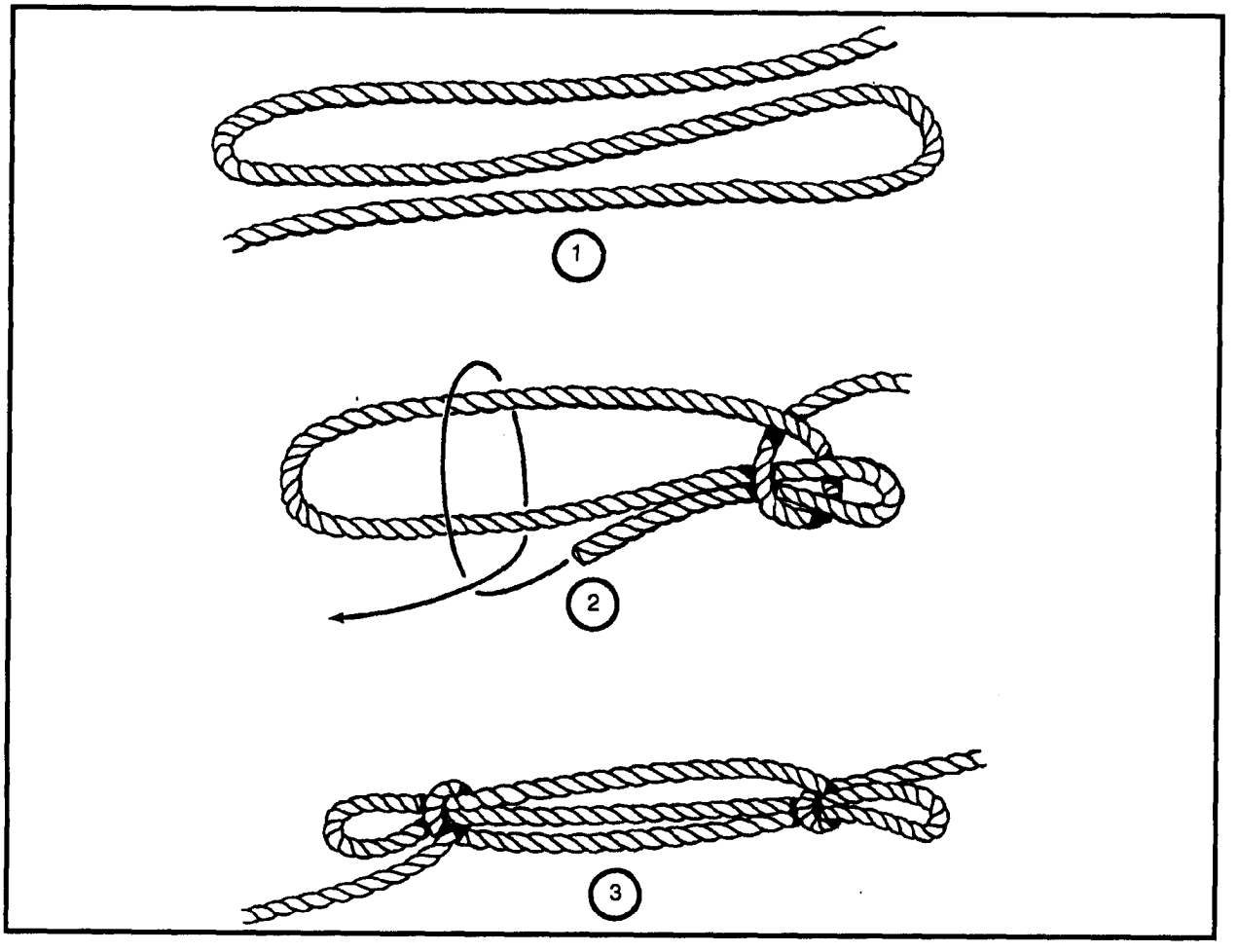


Figure 2-34. Sheepshank

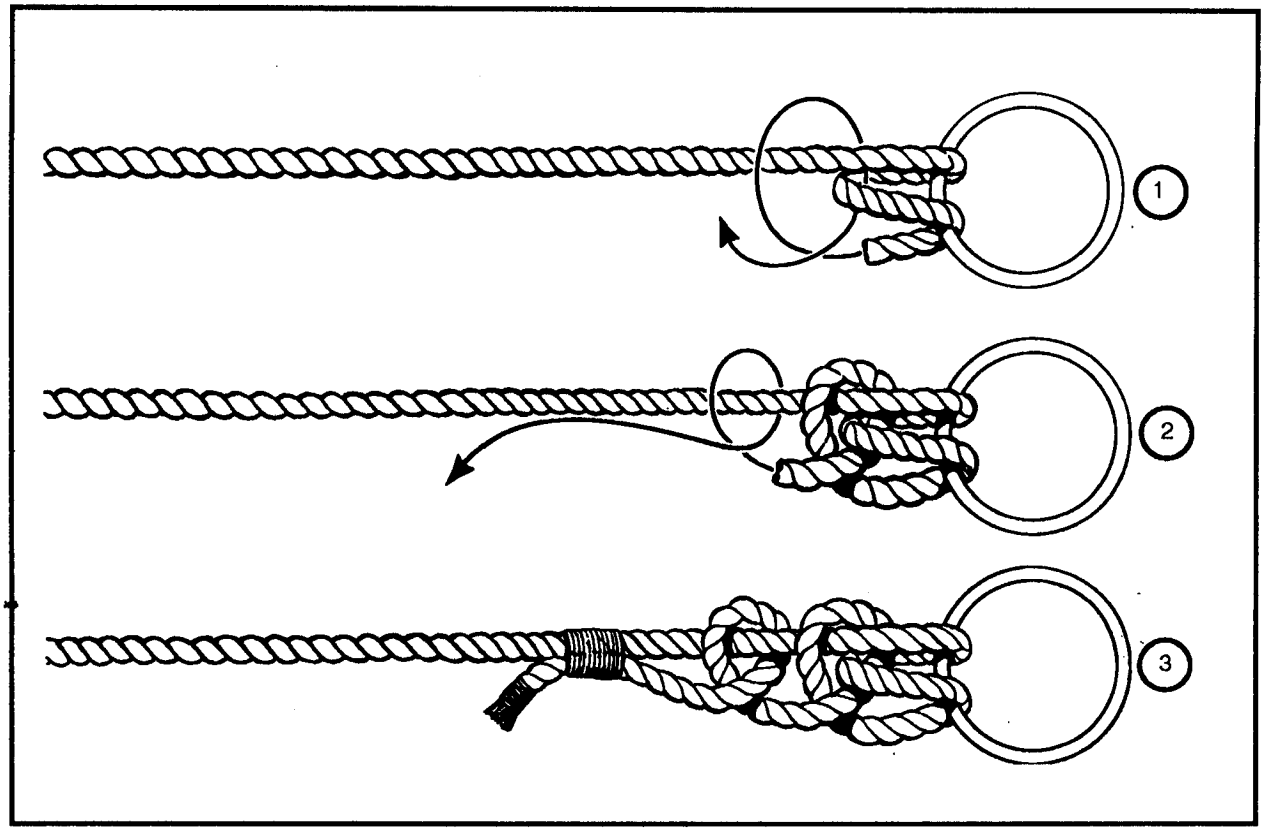


Figure 2-35. Fisherman's bend

LASHINGS

A lashing is as rope, wire, or chain used for binding, wrapping, or fastening. The types of lashings include square, shears, and block.

SQUARE LASHING

Use the square lashing to lash two spars together at right angles to each other (see *Figure 2-36*). To tie a square lashing, begin with a clove hitch on one spar and make a minimum of four complete turns around both members. Continue with two frapping turns between the vertical and the horizontal spar to tighten the lashing. Tie off the running end to the opposite spar from which you started with another clove hitch to finish the square lashing.

SHEARS LASHING

Use the shears lashing to lash two spars together at one end to form an expedient device called a shears (see *Figure 2-37*). Do this by laying two spars side by side, spaced about one-third of the diameter of a spar apart, with the butt ends together. Start the shears lashing a short distance in from the top of one of the spars by tying the end of the rope to it with a clove hitch. Then make eight tight turns around both spars above the clove hitch. Tighten the lashing with a minimum of two frapping turns around the eight turns. Finish the shears lashing by tying the end of the rope to the opposite spar from which you started with another clove hitch.